

2. Homogeneous Differential Equations

When the variables in a first-order differential equation cannot be separated directly (i.e., you cannot get all x 's on one side and y 's on the other), the next method to check is whether the equation is **Homogeneous**.

1. Concept Overview

1.1 What is a Homogeneous Function?

A function $F(x, y)$ is called homogeneous of degree n if, when you replace x with λx and y with λy , the parameter λ can be factored out completely as λ^n .

$$F(\lambda x, \lambda y) = \lambda^n F(x, y)$$

1.2 What is a Homogeneous Differential Equation?

A first-order differential equation $\frac{dy}{dx} = f(x, y)$ is homogeneous if the function $f(x, y)$ can be expressed as a ratio of two homogeneous functions of the **same degree**.

Alternatively, if you can write the differential equation in the form:

$$\frac{dy}{dx} = F\left(\frac{y}{x}\right)$$

then it is a Homogeneous Differential Equation.

2. Working Rule (The Substitution Method)

To solve a homogeneous differential equation, we use a specific substitution to transform it into a **Variable Separable** equation.

1. **Identify Homogeneity:** Check that every term in the numerator and denominator has the same total power of variables.
2. **Substitute:**
 - Let $y = vx$
 - Differentiate with respect to x (using the product rule):

$$\frac{dy}{dx} = v + x \frac{dv}{dx}$$

3. **Transform:** Replace y and $\frac{dy}{dx}$ in the original equation. The x terms will typically cancel out, leaving an equation in terms of v and x .
 4. **Separate and Integrate:** The new equation will be Variable Separable. Separate v and x , then integrate both sides.
 5. **Resubstitute:** Finally, replace v with $\frac{y}{x}$ to get the general solution in terms of x and y .
-

3. Solved Questions

Here are questions from the lecture notes demonstrating the homogeneity check and the solution process.

Question 1: Checking for Homogeneity

Determine if the function $F(x, y) = \frac{2x^2y}{x^2+y^2}$ is homogeneous.

Solution: 1. **Test:** Replace $x \rightarrow \lambda x$ and $y \rightarrow \lambda y$.

$$F(\lambda x, \lambda y) = \frac{2(\lambda x)^2(\lambda y)}{(\lambda x)^2 + (\lambda y)^2}$$

2. **Simplify:**

$$\begin{aligned} &= \frac{2\lambda^2 x^2 \cdot \lambda y}{\lambda^2 x^2 + \lambda^2 y^2} \\ &= \frac{\lambda^3 (2x^2 y)}{\lambda^2 (x^2 + y^2)} \end{aligned}$$

3. **Factor:**

$$= \lambda^{3-2} \left(\frac{2x^2 y}{x^2 + y^2} \right) = \lambda^1 F(x, y)$$

4. **Conclusion:** Yes, it is a homogeneous function of degree 1.

Question 2: Solving a Homogeneous DE

Solve: $(2xy + x^2) \frac{dy}{dx} = 3y^2 + 2xy$

Solution: 1. **Rewrite in standard form:**

$$\frac{dy}{dx} = \frac{3y^2 + 2xy}{2xy + x^2}$$

Observation: Every term (y^2, xy, x^2) is of degree 2. Thus, it is Homogeneous. 2. **Apply Substitution:** Let $y = vx$, so $\frac{dy}{dx} = v + x \frac{dv}{dx}$. Substitute into the equation:

$$v + x \frac{dv}{dx} = \frac{3(vx)^2 + 2x(vx)}{2x(vx) + x^2}$$

3. **Simplify the Fraction:**

$$\begin{aligned} v + x \frac{dv}{dx} &= \frac{x^2(3v^2 + 2v)}{x^2(2v + 1)} \\ v + x \frac{dv}{dx} &= \frac{3v^2 + 2v}{2v + 1} \end{aligned}$$

4. **Separate Variables:** Move v to the right side:

$$x \frac{dv}{dx} = \frac{3v^2 + 2v}{2v + 1} - v$$

Find a common denominator:

$$\begin{aligned} x \frac{dv}{dx} &= \frac{3v^2 + 2v - v(2v + 1)}{2v + 1} \\ x \frac{dv}{dx} &= \frac{3v^2 + 2v - 2v^2 - v}{2v + 1} = \frac{v^2 + v}{2v + 1} \end{aligned}$$

Now, separate v and x :

$$\frac{2v + 1}{v^2 + v} dv = \frac{1}{x} dx$$

5. **Integrate:**

$$\int \frac{2v + 1}{v^2 + v} dv = \int \frac{1}{x} dx$$

Notice: The numerator $(2v + 1)$ is the exact derivative of the denominator $(v^2 + v)$.

$$\ln |v^2 + v| = \ln |x| + \ln C$$

$$v^2 + v = Cx$$

6. **Resubstitute** $v = y/x$:

$$\left(\frac{y}{x}\right)^2 + \frac{y}{x} = Cx$$

$$\frac{y^2 + xy}{x^2} = Cx \implies y^2 + xy = Cx^3$$

Question 3: Trigonometric Homogeneous DE

Solve: $x \sin\left(\frac{y}{x}\right) dy = [y \sin\left(\frac{y}{x}\right) - x] dx$

Solution: 1. **Rewrite in standard form:**

$$\frac{dy}{dx} = \frac{y \sin(y/x) - x}{x \sin(y/x)}$$

$$\frac{dy}{dx} = \frac{y}{x} - \frac{x}{x \sin(y/x)} = \frac{y}{x} - \frac{1}{\sin(y/x)}$$

Observation: The equation is a function of (y/x) , so it is Homogeneous. 2. **Apply Substitution:** Let $y = vx$, so $\frac{dy}{dx} = v + x \frac{dv}{dx}$.

$$v + x \frac{dv}{dx} = v - \frac{1}{\sin v}$$

3. **Simplify:** Subtract v from both sides:

$$x \frac{dv}{dx} = -\frac{1}{\sin v} = -\csc v$$

4. **Separate Variables:**

$$\sin v dv = -\frac{1}{x} dx$$

5. **Integrate:**

$$\int \sin v dv = -\int \frac{1}{x} dx$$

$$-\cos v = -\ln |x| + C$$

Multiply by -1 :

$$\cos v = \ln |x| - C$$

6. **Resubstitute** $v = y/x$:

$$\cos\left(\frac{y}{x}\right) = \ln |x| + C'$$

Next Method: What if the equation looks homogeneous but has constant terms interfering? Proceed to [Method 3: Equations Reducible to Homogeneous Form](#).